Randomized and low-rank approximation

Optimal dimension for a rerandomized Hadamard embedding

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Rating rationale: Challenging because two structured randomizations must remove the embedding logarithm uniformly over subspaces; community impact is fast sketching for least squares and low-rank methods.

Problem statement

Let n be a power of two, $1 \le r \le n$, and $0 < \varepsilon < 1/2$. Let F be the normalized Walsh–Hadamard matrix, let D_1, D_2 have independent Rademacher diagonal entries, and let $S \in \mathbb{R}^{n\times k}$ select a uniformly random k-element subset of coordinates, independently. Set

$$\Omega = \sqrt{n/k}\, D_1 F D_2 F S.$$

Does a universal $C > 0$ exist such that, for every fixed r-dimensional subspace $V \subseteq \mathbb{R}^n$, choosing $k = \min\{n, \lceil Cr/\varepsilon^2 \rceil\}$ gives

$$\Pr\left\{(1-\varepsilon)\|x\|_2^2 \le \|\Omega^T x\|_2^2 \le (1+\varepsilon)\|x\|_2^2 \text{ for every } x \in V\right\} \ge 0.99?$$

This fixes normalization, sampling without replacement, and a constant success probability in the workshop question. Results for a single randomization, three randomizations, or independently sampled sparse sketches do not establish this statement.

References

Amsel et al., *Linear Systems and Eigenvalue Problems: Open Questions from a Simons Workshop*, §5.3, Definition 5.3 and Problem 5.6. For the single-round baseline, Joel A. Tropp, *Improved Analysis of the Subsampled Randomized Hadamard Transform*, *Advances in Adaptive Data Analysis* 3 (2011), 115–126, Theorem 3.1.

Status check

Searches included *"rerandomized SRHT" counterexample*, *"rerandomized" "subspace" Hadamard*, and *"Hadamard" "two" "2026" embedding conjecture*. No resolution of the displayed two-round assertion was located. The September 2026 SparseStack result in the screening notes concerns another distribution.

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Rechecked workshop Problem 5.6. The precise two-round Hadamard distribution remains unproved there. Rerandomized-Hadamard and later embedding searches found no resolution; guarantees for independent sparse entries or additional randomizations do not settle this distribution.